

Si–O–Zr bonded SiC@UiO-66-NH₂ aerogel for enhanced photocatalytic degradation of tetracycline via interfacial charge regulation

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ABSTRACT

Interfacial charge regulation is essential for improving photocatalytic antibiotic degradation, yet conventional powder photocatalysts still suffer from poor recovery and catalyst loss. Herein, a recyclable SiC@UiO-66-NH₂ aerogel with chemically bonded Si–O–Zr heterointerfaces was fabricated through surface hydroxylation of SiC nanofiber aerogels followed by the in situ growth of UiO-66-NH₂. The SiC aerogel acts as a robust monolithic framework for efficient mass transfer and convenient recovery, while UiO-66-NH₂ enriches tetracycline molecules near the active sites. The chemically bonded Si–O–Zr heterointerface enables strong electronic coupling between SiC and UiO-66-NH₂, promotes directional charge transfer, and suppresses photogenerated carrier recombination. Benefiting from these synergistic effects, SiC@UiO-66-NH₂ achieved 91.65% tetracycline degradation within 2 h under Xe lamp irradiation, with an apparent pseudo-first-order rate constant of 0.01516 min^{−1} over the 20–120 min fitting interval, exceeding pristine SiC, UiO-66-NH₂, and physically mixed SiC/UiO-66-NH₂. It also maintained high activity over pH 5–9, across tetracycline concentrations of 5–40 mg L^{−1}, and after 15 consecutive cycles with 98.2% mass retention. Time-resolved LC-MS and TOC analyses showed continued oxidation of intermediates and 56.78% TOC removal after 120 min, while scavenger tests identified ·OH as the predominant reactive species. Mechanistic analyses and density functional theory calculations confirm that the Si–O–Zr interface induces charge redistribution, generates a built-in electric field, narrows the band gap, and enhances tetracycline adsorption. This study highlights chemically bonded MOF/SiC aerogel heterostructures as efficient and recyclable photocatalysts for antibiotic-contaminated water remediation.

1. Introduction

Tetracycline (TC), as a broad-spectrum antibiotic, has been widely used in human medicine, livestock breeding, and aquaculture. [1] However, TC is difficult to completely metabolize, and its residues can continuously enter aquatic environments through pharmaceutical wastewater, agricultural runoff, and municipal sewage, thereby inducing antibiotic resistance and posing potential risks to ecosystems and human health. [2,3] Therefore, the efficient elimination of TC from contaminated water is of great significance for mitigating antibiotic-residue-related environmental risks. Photocatalysis, which can utilize light energy to generate reactive species for antibiotic oxidation and even mineralization, is regarded as a green and efficient strategy for water pollution remediation [4,5]. Nevertheless, conventional powder photocatalysts generally suffer from aggregation, difficult separation,

catalyst loss, and potential secondary pollution during practical operation. In addition, insufficient visible-light utilization, weak pollutant adsorption, and rapid recombination of photogenerated carriers further limit their degradation efficiency. Therefore, developing photocatalytic systems that combine high degradation activity, strong environmental adaptability, and low catalyst-loss risk is highly desirable for hazardous antibiotic wastewater treatment.

Immobilizing photocatalysts into monolithic porous architectures provides an effective route to reduce catalyst loss and secondary particulate contamination during water treatment [6,7]. Compared with dispersed powders, monolithic aerogels possess ultralight weight, high porosity, open mass-transfer channels, and easy solid–liquid separation, which can not only improve the contact efficiency between pollutants and active sites but also facilitate catalyst recovery and reuse [8]. For example, Liu et al. constructed highly compressible CNF/rGO/g-C₃N₄

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aerogels for visible-light photocatalysis, where the three-dimensional porous framework improved photocatalytic activity and enabled convenient reuse, with an RhB degradation efficiency of 87.3% and good mechanical stability after repeated compression [9]. Nevertheless, most reported photocatalytic aerogels rely on carbonaceous or polymeric frameworks, and their long-term structural robustness under complex aqueous environments remains a concern. Silicon carbide (SiC), with excellent chemical stability, thermal robustness, and corrosion resistance, can form a continuous three-dimensional nanofiber aerogel skeleton, providing a stable and recyclable platform for photocatalytic reactions [10–12]. However, pristine SiC shows limited adsorption capacity toward aromatic antibiotic molecules, and its photogenerated electron–hole pairs are prone to rapid recombination, resulting in unsatisfactory photocatalytic degradation efficiency [13]. Therefore, functional component coupling and interfacial regulation are required to further enhance the photocatalytic performance and environmental applicability of SiC aerogels.

Constructing heterojunctions is an important strategy for improving semiconductor photocatalysis, because a rational heterointerface can promote the spatial separation of photogenerated electrons and holes, suppress carrier recombination, and strengthen the redox capability of the system. [14–17] UiO-66-NH₂, a typical amino-functionalized Zr-based metal–organic framework, possesses a high specific surface area, stable porous structure, and visible-light response. [18,19] Its porous framework and –NH₂ groups can enrich TC molecules through hydrogen bonding, electrostatic interaction, or π – π interaction, forming a local high-concentration microenvironment favorable for photocatalytic reactions. Ren et al. reported a (Zr/Ce)UiO-66(NH₂)@g-C₃N₄ Z-scheme heterojunction for TC photodegradation, in which the MOF-mediated charge transfer and interfacial Z-scheme pathway significantly enhanced the degradation kinetics, with the rate constant being 23.73 times higher than that of UiO-66 [20]. However, such powder heterojunction systems still face separation and recovery challenges, and weak physical contact between components may limit stable interfacial charge transfer. [21,22] Therefore, in situ growth of UiO-66-NH₂ on a robust SiC aerogel skeleton and the construction of a chemically bonded heterointerface are expected to simultaneously realize pollutant adsorption enrichment, efficient photogenerated charge separation, and monolithic catalyst recovery.

Herein, we propose an interfacial engineering strategy to construct chemically bonded SiC@UiO-66-NH₂ composite aerogels by activating SiC nanofiber aerogels through surface hydroxylation, followed by the in situ solvothermal growth of UiO-66-NH₂ on the SiC surface. To elucidate the role of the chemically bonded heterointerface, pristine SiC aerogels, UiO-66-NH₂ powder, and physically mixed SiC/UiO-66-NH₂ were used as comparison samples. The resulting SiC@UiO-66-NH₂ exhibits markedly enhanced photocatalytic performance toward TC degradation, achieving 91.65% removal within 2 h under Xe lamp irradiation and outperforming all comparison samples. In addition to its high activity, the composite aerogel demonstrates excellent recyclability, maintaining over 90% degradation efficiency after 15 cycles with negligible catalyst loss. Notably, it also shows strong environmental adaptability, sustaining high degradation efficiencies over a wide pH range and under different TC concentrations. Mechanistic studies reveal that the three-dimensional SiC aerogel framework provides a recyclable porous platform with efficient mass transfer, while UiO-66-NH₂ enhances TC adsorption and enrichment. More importantly, the in situ formed Si–O–Zr heterointerface promotes interfacial charge transfer and suppresses electron–hole recombination, thereby improving photocatalytic efficiency. This study demonstrates that chemically bonded MOF/ceramic aerogel heterostructures offer a promising strategy for the efficient, stable, and recyclable removal of antibiotic contaminants from water.

2. Materials and methods

2.1. Materials

Melamine foam (MF), silicon dioxide powder, silicon powder, nitric acid (HNO₃), zirconium tetrachloride (ZrCl₄), 2-aminoterephthalic acid (NH₂-BDC), *N,N*-dimethylformamide (DMF), acetic acid, and methanol were used for material preparation. All reagents were of analytical grade and used as received without further purification. Deionized water was used throughout the experiments.

2.2. Preparation of SiC aerogels

SiC aerogels were prepared by a chemical vapor deposition (CVD) method. First, melamine foam was carbonized at 973 K for 2 h under an Ar atmosphere to obtain carbon foam, which served as the carbon source for SiC nanofiber growth. Silicon dioxide and silicon powder were mixed at a molar ratio of 1:1 by ball milling at 300 rpm for 5 h under Ar atmosphere to obtain the silicon source. The mixed powder was placed in a cylindrical crucible, and the carbon foam was positioned above the powder. The CVD reaction was then carried out at 1723 K for 4 h under Ar atmosphere to obtain three-dimensional SiC nanofiber aerogels.

2.3. Preparation of SiC@UiO-66-NH₂ aerogels

To promote the in-situ nucleation of UiO-66-NH₂ and enhance interfacial bonding, the as-prepared SiC aerogels were first immersed in 5 M HNO₃ solution and treated at 80 °C for 4 h. Afterward, the aerogels were repeatedly washed with deionized water until neutral pH was reached and then dried under vacuum at 60 °C overnight.

For the solvothermal growth of UiO-66-NH₂, 0.233 g of ZrCl₄ was dissolved in 20 mL of DMF, while 0.181 g of NH₂-BDC was dissolved in another 20 mL of DMF under ultrasonication. The two solutions were then mixed, followed by the addition of 2–5 mL of acetic acid as a modulator. The hydroxylated SiC aerogel was placed in a Teflon-lined autoclave, and the precursor solution was added to fully immerse the aerogel. The autoclave was sealed and heated at 120 °C for 24 h. After naturally cooling to room temperature, the obtained composite aerogel was washed with fresh DMF twice for 2 h each, followed by methanol washing three times to remove residual reactants and replace DMF in the pores. Finally, the sample was dried under vacuum at 80 °C for 12 h to obtain the SiC@UiO-66-NH₂ composite aerogel.

2.4. Preparation of control samples

The SiC aerogel was obtained directly from the above CVD process. UiO-66-NH₂ powder was synthesized using the same solvothermal system without adding SiC aerogel. The physically mixed SiC/UiO-66-NH₂ sample was prepared by grinding SiC aerogel with UiO-66-NH₂ powder and used as a control sample without a chemically bonded heterointerface.

2.5. Photocatalytic degradation experiments

The photocatalytic degradation performance of the prepared samples was evaluated using tetracycline (TC) as the target pollutant. Typically, the catalyst was added into 20.0 mg L^{−1} TC aqueous solution, and the catalyst dosage was fixed at 0.2 g L^{−1}. Before irradiation, the suspension was magnetically stirred in the dark to establish adsorption–desorption equilibrium. The photocatalytic reaction was then carried out under a 300 W Xe lamp. At given time intervals, aliquots of the suspension were collected and filtered to remove the catalyst. The residual TC concentration was quantified by high-performance liquid chromatography (HPLC, Agilent 1260).

The degradation efficiency was calculated according to the following equation [23]:

$$\eta = (1 - C/C_0) \times 100\%$$

where C_0 and C represent the TC concentrations after dark adsorption equilibrium and at irradiation time t , respectively. For comparison, SiC aerogel, UiO-66-NH₂ powder, physically mixed SiC/UiO-66-NH₂, and SiC@UiO-66-NH₂ aerogel were tested under identical conditions. To evaluate the cycling stability, the SiC@UiO-66-NH₂ aerogel was recovered after each run, washed, dried, and reused for the next cycle. The cycling test was conducted for 15 consecutive runs.

Photocatalytic experiments were conducted in a cubic glass photoreactor with dimensions of 50 mm × 50 mm × 50 mm, containing 50 mL of reaction solution. For the monolithic SiC@UiO-66-NH₂ aerogel, the sample was fixed with a thin electrode clip and vertically positioned at the center of the solution, with its surface directly facing the incident light. The electrode clip was maintained above the liquid level, and only the aerogel was immersed in the solution. A 300 W Xe lamp was positioned at the side of the reactor to provide horizontal irradiation, with a fixed distance of 20 cm between the lamp and the photoreactor. The emission spectrum of the Xe lamp is provided in Fig. S1. The temperature of the reaction solution was monitored throughout irradiation and remained relatively stable, increasing only slightly from 25.0 °C to 27.5 °C during the 120 min experiment, with a maximum temperature of 27.7 °C (Fig. S2). No external oxygen bubbling or aeration was applied during the photocatalytic experiments, which were conducted under ambient air with continuous stirring.

Photolysis blank experiments were performed without catalyst under otherwise identical Xe-lamp irradiation for 140 min. Extended dark-adsorption experiments were conducted for UiO-66-NH₂, SiC, SiC@UiO-66-NH₂, and physically mixed SiC/UiO-66-NH₂ for 140 min using an initial TC concentration of 20 mg L⁻¹ and a catalyst dosage of 0.2 g L⁻¹. Scavenger experiments were performed by adding p-benzoquinone (BQ), isopropanol (IPA), or ammonium oxalate (AO) as scavengers for ·O₂⁻, ·OH, or h⁺, respectively.

2.6. Characterization

The morphology and microstructure of the samples were characterized by scanning electron microscopy (SEM), transmission electron microscopy (TEM), high-resolution TEM (HRTEM), and energy-dispersive X-ray spectroscopy (EDS) mapping. The crystal structures were analyzed by X-ray diffraction (XRD). X-ray photoelectron spectroscopy (XPS) was used to investigate the surface elemental composition and chemical bonding states of the samples. UV-vis diffuse reflectance spectroscopy (UV-vis DRS) was employed to evaluate the optical absorption properties, and the band gaps were estimated from the corresponding Tauc plots. Photoluminescence (PL) spectroscopy was used to analyze the recombination behavior of photogenerated electron-hole pairs. High-resolution O 1s XPS spectra were analyzed to further resolve the oxygen coordination environments. N₂ adsorption-desorption measurements were performed for all four samples; multipoint BET analysis was used to determine the specific surface areas, while NLDFT adsorption-branch analysis in the 1–5 nm range and BJH adsorption-branch analysis in the 5–20 nm range were used to evaluate the pore-size distributions. Inductively coupled plasma (ICP) analysis was used to determine the Zr content of SiC@UiO-66-NH₂ after 120 min photocatalytic treatment at pH 3, 5, 7, 9, and 11, and the treated samples were further examined by SEM and XRD to evaluate morphological and structural stability.

2.7. Photoelectrochemical measurements

Photoelectrochemical measurements were performed to investigate the separation and transfer behavior of photogenerated charges. Transient photocurrent responses were recorded under intermittent light irradiation. Electrochemical impedance spectroscopy (EIS) was conducted to evaluate the interfacial charge-transfer resistance. All samples

were tested under the same conditions for comparison.

2.8. LC-MS analysis

Liquid chromatography-mass spectrometry (LC-MS) was used to identify and track the intermediates generated during TC degradation over SiC@UiO-66-NH₂. [24] Reaction solutions were collected at 0, 20, 40, 60, 90, and 120 min, filtered, and analyzed by LC-MS. The TC peak area was normalized to its value at 0 min, whereas each intermediate was normalized to its maximum peak area during the reaction. Total organic carbon (TOC) measurements were conducted at 0, 20, 40, 60, 80, and 120 min to evaluate the mineralization behavior. Based on the detected intermediates and reported TC degradation pathways, the possible degradation pathway of TC over SiC@UiO-66-NH₂ was proposed.

2.9. Density functional theory calculations

First-principles calculations based on density functional theory (DFT) were performed using the Vienna Ab initio Simulation Package (VASP) [25]. The generalized gradient approximation (GGA) with the Perdew-Burke-Ernzerhof (PBE) functional was employed to describe the exchange-correlation energy. The projected augmented wave (PAW) method was used, and the plane-wave cutoff energy was set to 400 eV. The convergence criteria for energy and force were set to 1 × 10⁻⁵ eV and 1 × 10⁻² eV Å⁻¹, respectively. A vacuum layer of 15 Å was introduced along the z direction to avoid interactions between periodically repeated slabs. The Brillouin zone was sampled using the Monkhorst-Pack scheme, and the k -point mesh was generated by VASPKIT with a recommended value of 0.04 (2π × 0.04 Å⁻¹) [26]. Interatomic charge transfer was analyzed by Bader charge analysis.

The charge density difference was calculated to visualize the interfacial electron redistribution in the SiC@UiO-66-NH₂ heterostructure. The density of states (DOS) was analyzed to reveal the electronic structure and band-gap variation after heterostructure formation. The electrostatic potential and work function were calculated to clarify the charge-transfer direction and the formation of the internal electric field. In addition, the adsorption energies of TC on SiC, UiO-66-NH₂, and SiC@UiO-66-NH₂ were calculated to compare their adsorption affinity toward TC molecules.

3. Results and discussion

The fabrication process of the SiC@UiO-66-NH₂ composite aerogel is schematically illustrated in Fig. 1a. A three-dimensional SiC nanofiber aerogel was first prepared by a CVD method, followed by surface hydroxylation to introduce anchoring sites for UiO-66-NH₂ nucleation. Subsequently, UiO-66-NH₂ nanoparticles were grown in situ on the SiC nanofibers through a solvothermal process, yielding SiC@UiO-66-NH₂ composite aerogels with intimate heterointerfaces. This in situ growth strategy is expected to suppress the random aggregation of UiO-66-NH₂ particles and strengthen the interfacial contact between UiO-66-NH₂ and the SiC skeleton, which is essential for efficient interfacial charge transfer during photocatalysis.

The morphology of pristine SiC aerogel is shown in Fig. 1b. The aerogel consists of abundant interwoven nanofibers with diameters of approximately 50 nm and lengths of tens of micrometers, forming a continuous three-dimensional porous network. Such an open nanofiber framework is favorable for solution penetration, pollutant diffusion, and catalyst recovery. After the in-situ growth of UiO-66-NH₂, the SiC@UiO-66-NH₂ aerogel retains the integrated fibrous network, while numerous UiO-66-NH₂ nanoparticles with a size of about 200 nm are uniformly anchored on the surface of SiC nanofibers (Fig. 1c). These particles are distributed along the SiC fibers in a bead-like manner, indicating that UiO-66-NH₂ preferentially nucleates and grows on the activated SiC surface rather than randomly depositing within the aerogel pores. In

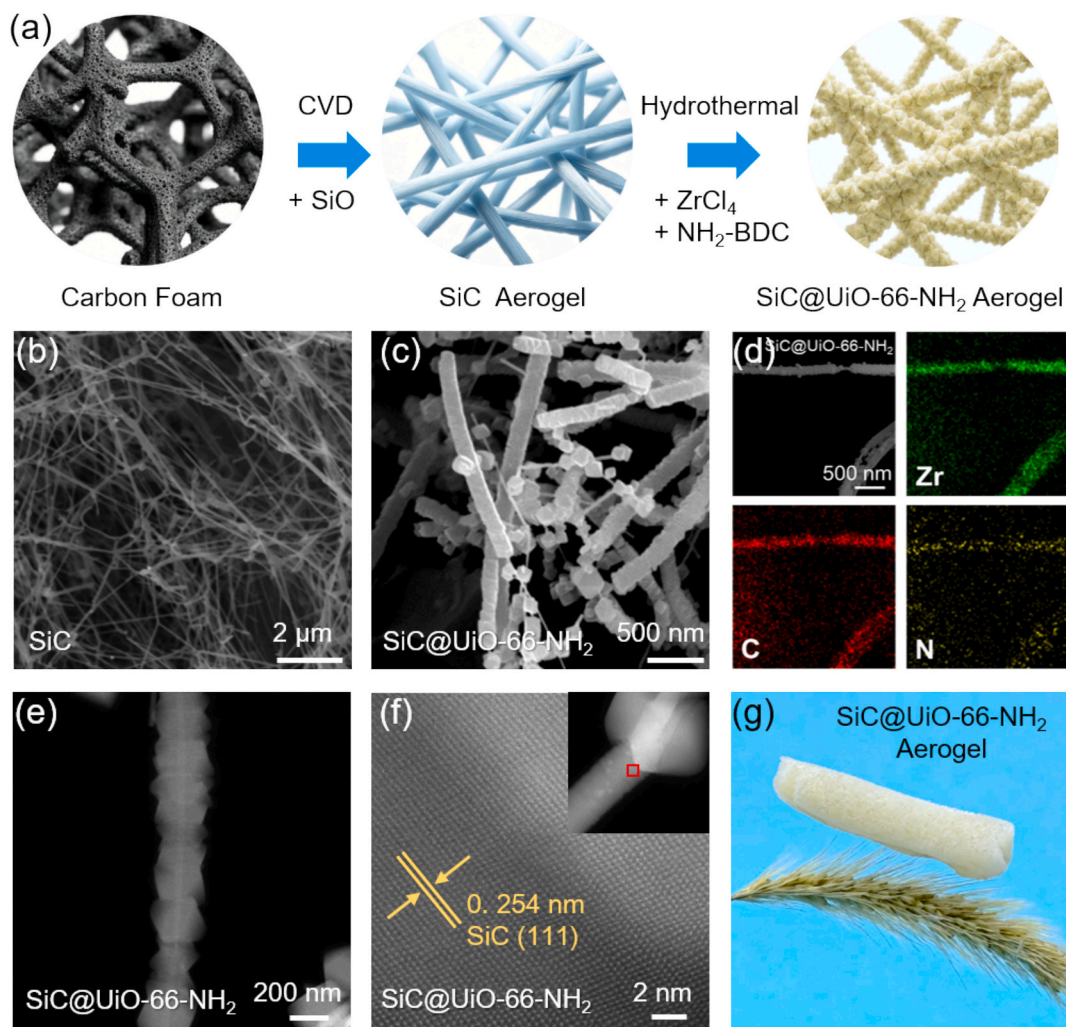


Fig. 1. (a) Schematic illustration of the fabrication process of SiC@UiO-66-NH₂ aerogels. (b) SEM image of pristine SiC aerogel. (c) SEM image of SiC@UiO-66-NH₂ aerogel. (d) SEM image and corresponding EDS elemental mapping of a single SiC@UiO-66-NH₂ composite fiber. (e) TEM and (f) HRTEM images of the SiC@UiO-66-NH₂. (g) Optical photograph of the ultralight SiC@UiO-66-NH₂ aerogel supported on a *setaria viridis*.

contrast, the physically mixed SiC/UiO-66-NH₂ sample exhibits loose and random contact between SiC fibers and UiO-66-NH₂ particles (Fig. S3), further confirming the structural advantage of the in-situ growth strategy.

SEM-EDS mapping of a single SiC@UiO-66-NH₂ composite fiber reveals the uniform distribution of Zr, C, and N elements along the fiber (Fig. 1d), confirming the homogeneous loading of UiO-66-NH₂ on the SiC surface. The TEM image in Fig. 1e further reveals a core-shell-like SiC@UiO-66-NH₂ architecture, where the SiC nanofiber serves as the core and UiO-66-NH₂ nanoparticles are uniformly distributed around and coated along the fiber, consistent with the bead-like growth behavior observed in the SEM images. The HRTEM image in Fig. 1f displays a clear lattice spacing of 0.254 nm, corresponding to the (111) crystal plane of SiC, further confirming the crystalline structure of the SiC component in the composite. The optical photograph in Fig. 1g shows that the obtained SiC@UiO-66-NH₂ aerogel is light yellow and can be steadily supported on *Setaria viridis*, demonstrating its ultralight and monolithic characteristics. Benefiting from the continuous porous skeleton and intimate SiC/UiO-66-NH₂ heterointerface, the composite aerogel is expected to provide accessible active sites, rapid mass-transfer channels, and convenient recyclability for photocatalytic TC degradation.

XRD analysis was performed to further confirm the structural composition of the SiC@UiO-66-NH₂ composite aerogel. As shown in

Fig. 2a, the pristine SiC aerogel and UiO-66-NH₂ powder exhibit their characteristic diffraction peaks, indicating the successful preparation of both individual components [11]. The experimental XRD pattern of UiO-66-NH₂ is in good agreement with the simulated pattern (Fig. S4a), confirming the well-defined crystalline structure of the obtained UiO-66-NH₂. For the SiC@UiO-66-NH₂ composite aerogel, the characteristic diffraction peaks of both SiC and UiO-66-NH₂ can be observed, suggesting that UiO-66-NH₂ retains its crystalline structure after in situ growth on the SiC aerogel and that the SiC skeleton remains intact. In comparison, the physically mixed SiC/UiO-66-NH₂ sample also displays the characteristic peaks of both components (Fig. S4b), indicating that XRD mainly reflects the phase composition and cannot directly distinguish the interfacial difference between in situ growth and physical mixing. Therefore, XPS analysis was further conducted to investigate the surface chemical states and interfacial bonding.

The XPS survey spectrum (Fig. S5) shows the coexistence of Si, Zr, C, N, and O elements in SiC@UiO-66-NH₂, further confirming the successful incorporation of UiO-66-NH₂ into the SiC aerogel. To clarify whether a chemically bonded heterointerface was formed between SiC and UiO-66-NH₂, high-resolution Si 2p and Zr 3d XPS spectra were analyzed. As shown in Fig. 2c, the peaks located at approximately 100.5 and 103.5 eV in pristine SiC can be assigned to Si-C and Si-O bonds, respectively [13]. In contrast, a new Si 2p peak appears at 101.6 eV for SiC@UiO-66-NH₂, which can be attributed to the formation of Si-O-Zr

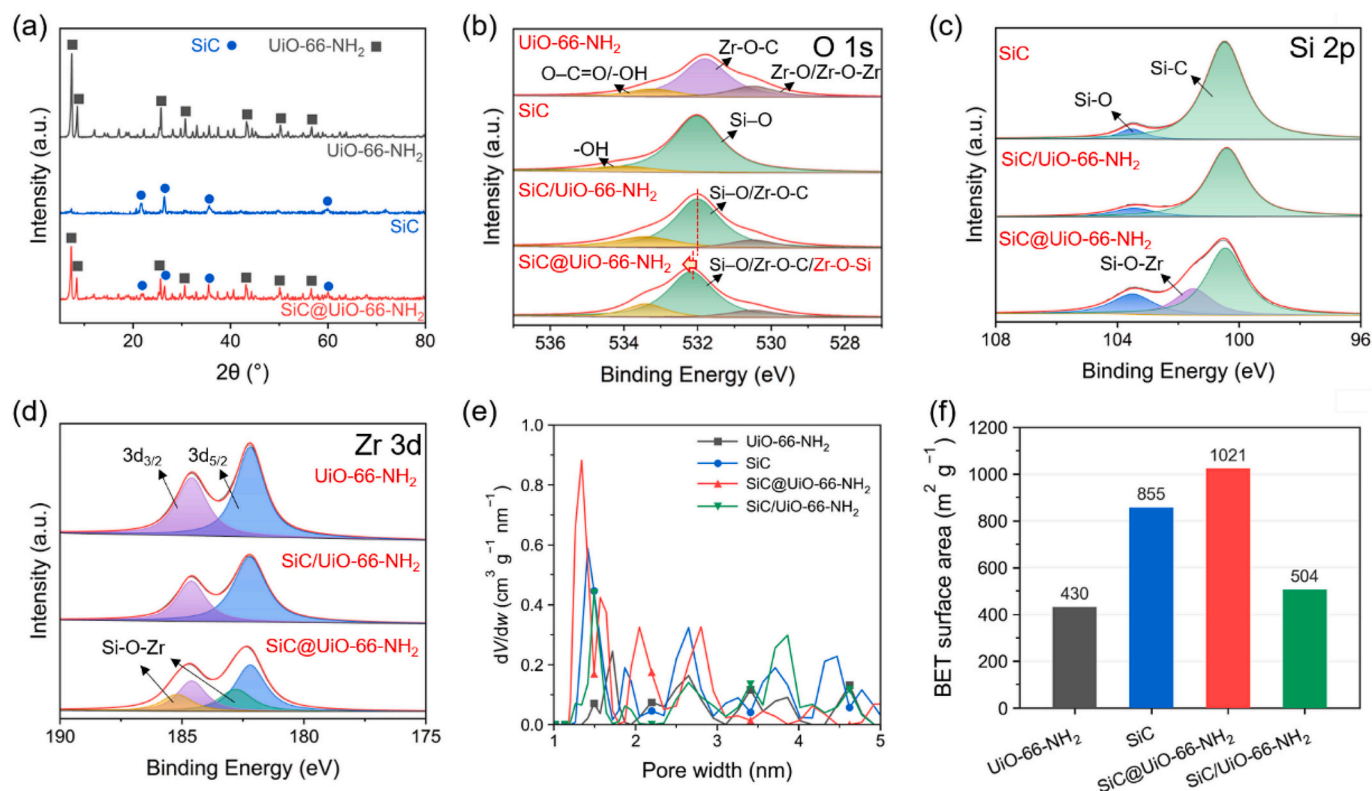


Fig. 2. (a) XRD patterns of SiC@UiO-66-NH₂, SiC, and UiO-66-NH₂. (b) High-resolution O 1s XPS of SiC@UiO-66-NH₂, SiC, SiC/UiO-66-NH₂, and UiO-66-NH₂. (c) High-resolution Si 2p XPS spectra of SiC@UiO-66-NH₂, SiC, and SiC/UiO-66-NH₂. (d) High-resolution Zr 3d XPS spectra of SiC@UiO-66-NH₂, SiC/UiO-66-NH₂, and UiO-66-NH₂. (e) Pore-size distributions of the four samples determined by NLDFT adsorption-branch analysis and (f) BET specific surface areas of UiO-66-NH₂, SiC, SiC@UiO-66-NH₂, and SiC/UiO-66-NH₂.

bonds, indicating chemical bonding between UiO-66-NH₂ and the hydroxylated SiC surface. Notably, this new peak is absent in the physically mixed SiC/UiO-66-NH₂ sample, suggesting that simple grinding cannot effectively construct a Si-O-Zr chemical interface.

High-resolution O 1s XPS spectra were further analyzed to clarify the interfacial oxygen environment (Fig. 2b). The O 1s spectrum of UiO-66-NH₂ can be deconvoluted into components centered at approximately 530.5, 531.8, and 533.2 eV, which are assigned to Zr-O/Zr-O-Zr, Zr-O-C, and O-C=O/-OH species, respectively [27]. For pristine SiC, the main O 1s components at approximately 532.0 and 534.0 eV are attributed to Si-O and surface -OH species. In the physically mixed SiC/UiO-66-NH₂ sample, the dominant component at approximately 532.0 eV mainly reflects overlapping Si-O and Zr-O-C contributions.

In contrast, the dominant O 1s component of SiC@UiO-66-NH₂ is further shifted to approximately 532.2 eV, where contributions from Si-O, Zr-O-C, and the interfacial Zr-O-Si environment are involved. Compared with the physically mixed sample, this additional positive shift indicates that the oxygen coordination environment is altered by interfacial chemical coupling rather than by simple spectral superposition. Meanwhile, the Zr-O/Zr-O-Zr component remains nearly unchanged at approximately 530.5 eV, suggesting that the major variation is localized in the interface-related oxygen environment. Together with the Si 2p and Zr 3d results, the O 1s analysis provides further support for a chemically coupled Si-O-Zr interface.

The high-resolution Zr 3d spectra further support this conclusion. As shown in Fig. 2d, the Zr 3d_{5/2} and Zr 3d_{3/2} peaks of UiO-66-NH₂ are located at approximately 182.2 and 184.6 eV, respectively [28]. In contrast, new peaks at around 182.8 and 185.2 eV are observed for SiC@UiO-66-NH₂, indicating a change in the local coordination environment of Zr, which can be attributed to the formation of Si-O-Zr bonds. By contrast, the physically mixed SiC/UiO-66-NH₂ sample does not show a similar obvious chemical shift, further confirming its loose

physical contact. These XRD and XPS results demonstrate that SiC@UiO-66-NH₂ not only successfully integrates UiO-66-NH₂ with the SiC aerogel, but also forms a chemically bonded heterointerface distinct from that in the physically mixed sample. This Si-O-Zr interface provides a crucial structural basis for efficient interfacial charge transfer and heterojunction effects in the subsequent photocatalytic process.

The textural properties of the four samples were evaluated by N₂ adsorption-desorption measurements. [29] The corresponding isotherms and multipoint BET linear fitting plots are shown in Figs. S6 and S7, respectively. The BET specific surface areas of UiO-66-NH₂, SiC, SiC@UiO-66-NH₂, and SiC/UiO-66-NH₂ are approximately 430, 855, 1021, and 504 m² g⁻¹, respectively (Fig. 2f), following the order SiC@UiO-66-NH₂ > SiC > SiC/UiO-66-NH₂ > UiO-66-NH₂.

NLDFT adsorption-branch analysis in the 1–5 nm region shows major pore-size peaks at approximately 1.34, 1.42, 1.49, and 1.72 nm for SiC@UiO-66-NH₂, SiC, SiC/UiO-66-NH₂, and UiO-66-NH₂, respectively (Fig. 2e), with SiC@UiO-66-NH₂ exhibiting a more pronounced contribution in this narrow-pore region. [30] In the 5–20 nm range, BJH adsorption-branch analysis indicates that UiO-66-NH₂, SiC, and SiC/UiO-66-NH₂ mainly exhibit pore-size contributions around 5–7 nm, whereas SiC@UiO-66-NH₂ shows a broader mesopore distribution with a local feature near 8.8 nm (Fig. S8). [31] These results provide direct experimental evidence for the distinct surface-area and pore-structure characteristics of the four materials.

Control experiments were performed before evaluating photocatalytic activity. In the absence of catalyst, the C/C₀ value remained close to 1.0 throughout 140 min of Xe-lamp irradiation and was 0.9982 at 140 min (Fig. S9), confirming negligible direct photolysis. During extended dark adsorption, all four samples approached equilibrium after approximately 80–100 min (Fig. S10). At 140 min, the C/C₀ values were 0.9504, 0.9703, 0.94547, and 0.9608 for UiO-66-NH₂, SiC, SiC@UiO-66-NH₂, and physically mixed SiC/UiO-66-NH₂, respectively,

corresponding to equilibrium adsorption capacities of 4.96, 2.97, 5.45, and 3.92 mg g⁻¹. Thus, direct photolysis contributes negligibly to TC removal, while SiC@UiO-66-NH₂ exhibits the highest experimental adsorption capacity.

Tetracycline (TC) was selected as the target pollutant to evaluate the photocatalytic degradation performance of the prepared samples under 300 W Xe lamp irradiation. The HPLC-derived degradation profiles are shown in Fig. 3a. Under identical reaction conditions ([TC]₀ = 20.0 mg L⁻¹, [catalyst] = 0.2 g L⁻¹), the C/C₀ values after 120 min were 0.436, 0.6299, 0.0835, and 0.5159 for UiO-66-NH₂, SiC, SiC@UiO-66-NH₂, and physically mixed SiC/UiO-66-NH₂, respectively, corresponding to TC degradation efficiencies of 56.40%, 37.01%, 91.65%, and 48.41%. SiC@UiO-66-NH₂ therefore exhibited the highest activity, whereas the physically mixed sample showed only limited enhancement, demonstrating that simple physical mixing is insufficient to establish an efficient charge-transfer pathway.

The HPLC-derived photocatalytic kinetic data were reanalyzed using

the pseudo-first-order relationship, $\ln(C_0/C) = k_{app}t$ (Fig. 3b) [32,33]. Because the initial 0–20 min stage showed a more rapid change and did not follow the same linear trend as the subsequent reaction period, linear fitting was performed over 20–120 min. A single apparent rate constant was obtained for each catalyst rather than calculating time-dependent rate constants. The resulting k_{app} values for UiO-66-NH₂, SiC, SiC@UiO-66-NH₂, and physically mixed SiC/UiO-66-NH₂ were 0.00365, 0.00314, 0.01516, and 0.00373 min⁻¹, respectively. The substantially larger k_{app} of SiC@UiO-66-NH₂ confirms its accelerated degradation kinetics under a consistent fitting treatment.

The stability and reusability of SiC@UiO-66-NH₂ were then evaluated by cycling experiments (Fig. 3c). The composite aerogel retained a TC degradation efficiency of over 90% after 15 consecutive cycles, indicating negligible activity decay. In addition, the mass retention of SiC@UiO-66-NH₂ remained above 98.2% after 15 cycles (Fig. S11), indicating that the monolithic aerogel structure effectively reduced catalyst loss during repeated recovery and reuse.

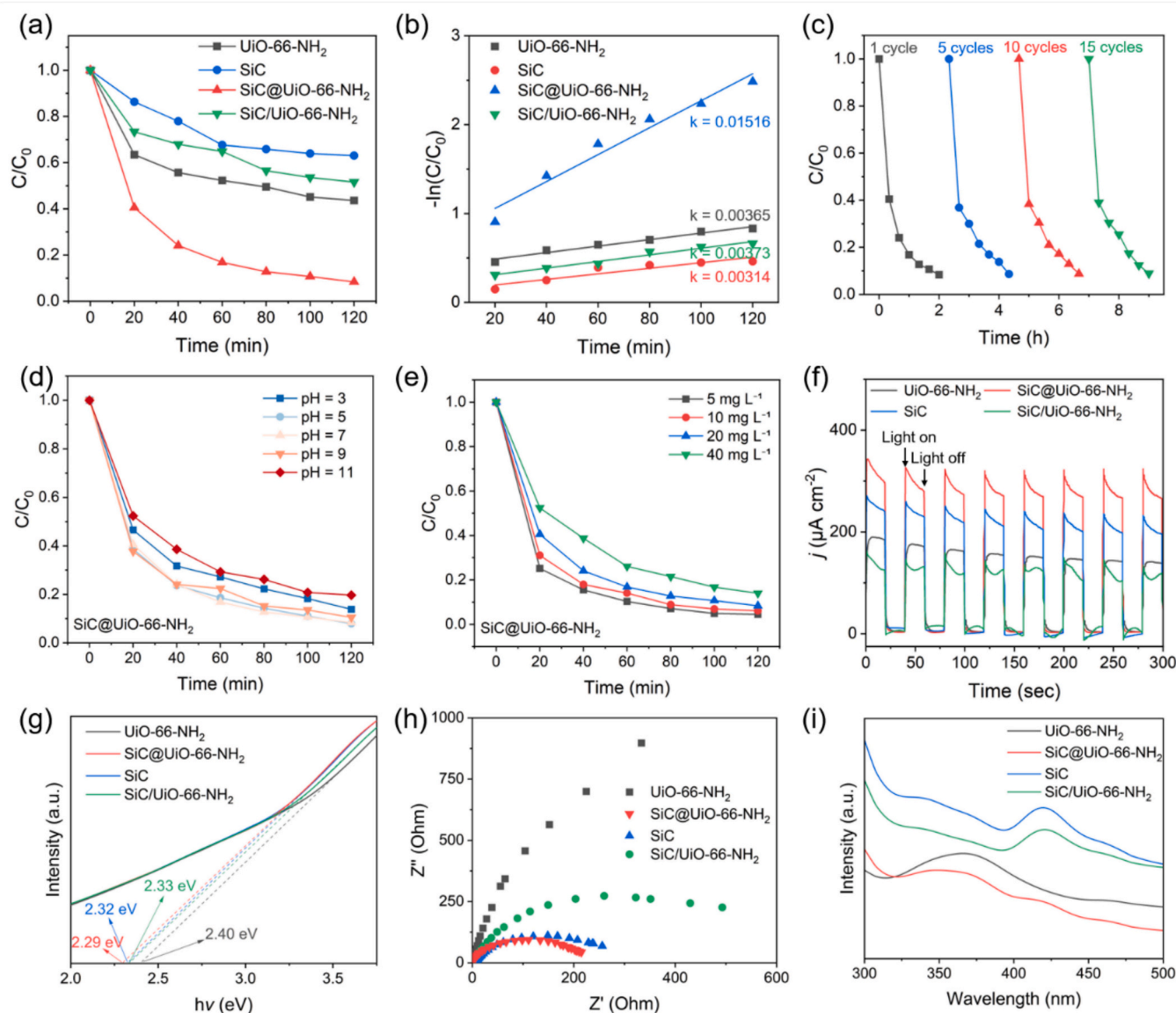


Fig. 3. (a) Photocatalytic degradation of TC over SiC@UiO-66-NH₂, UiO-66-NH₂, SiC, and SiC/UiO-66-NH₂. Reaction conditions: [TC]₀ = 20.0 mg L⁻¹, [catalyst] = 0.2 g L⁻¹. (b) Pseudo-first-order kinetic plots and linear fitting of TC degradation. (c) Cycling stability of SiC@UiO-66-NH₂ for TC degradation over 15 consecutive cycles. (d) Effect of initial solution pH on the photocatalytic degradation of TC over SiC@UiO-66-NH₂. (e) Effect of initial TC concentration on the photocatalytic degradation performance of SiC@UiO-66-NH₂. (f) Transient photocurrent responses, (g) Tauc plots derived from UV–Vis DRS spectra, (h) electrochemical impedance spectra, and (i) PL spectra of SiC@UiO-66-NH₂, UiO-66-NH₂, SiC, and SiC/UiO-66-NH₂.

To further evaluate the environmental adaptability of SiC@UiO-66-NH₂, TC degradation was investigated under different initial pH values (Fig. 3d). The degradation efficiency remained above 90% in the pH range of 5–9, suggesting that SiC@UiO-66-NH₂ can maintain efficient photocatalytic activity under weakly acidic, neutral, and weakly alkaline conditions. Even under strongly acidic and alkaline conditions, the degradation efficiencies still exceeded 85% at pH 3 and 80% at pH 11, respectively, demonstrating good pH tolerance. The effect of initial TC concentration was also examined (Fig. 3e). SiC@UiO-66-NH₂ achieved degradation efficiencies above 85% over a wide TC concentration range of 5–40 mg L⁻¹, indicating its favorable adaptability to different pollutant loads. These results demonstrate the excellent photocatalytic activity, rapid degradation kinetics, cycling stability, and environmental robustness of SiC@UiO-66-NH₂ for TC-contaminated water treatment.

The immobilization and structural stability of SiC@UiO-66-NH₂ were further assessed after 120 min photocatalytic treatment under different pH conditions. The Zr contents at pH 3, 5, 7, 9, and 11 were 14.55, 14.31, 14.56, 14.62, and 14.21 wt%, respectively (Fig. S12), remaining nearly unchanged across the tested pH range. SEM images after treatment at pH 3 and pH 11 and after 15 cycles show no obvious morphological deterioration (Fig. S13), while the corresponding XRD patterns remain consistent with that of the fresh sample without evident peak-position changes (Fig. S14). These results indicate that UiO-66-NH₂ is strongly immobilized on the SiC framework and that no obvious pH-dependent leaching or structural degradation occurs.

To reveal the origin of the enhanced photocatalytic performance of SiC@UiO-66-NH₂, photoelectrochemical measurements and spectroscopic analyses were performed to evaluate the light absorption ability, charge separation efficiency, interfacial charge transfer kinetics, and carrier recombination behavior of the prepared samples. Transient photocurrent response is an effective indicator of the generation, separation, and transport efficiency of photogenerated charge carriers under light irradiation. As shown in Fig. 3f, all samples exhibit reproducible photocurrent responses during repeated light on and off cycles, confirming their photoresponsive behavior. Notably, SiC@UiO-66-NH₂ delivers a much higher photocurrent density than SiC, SiC/UiO-66-NH₂, and UiO-66-NH₂, indicating that the in situ constructed composite aerogel can more efficiently separate and extract photogenerated electrons and holes. The relatively weak photocurrent response of the physically mixed SiC/UiO-66-NH₂ further suggests that simple mechanical contact is insufficient to establish an efficient charge transport pathway, highlighting the importance of the chemically bonded heterointerface.

The optical absorption properties of the samples were further investigated by UV–vis DRS, and the corresponding band gaps were estimated from the Tauc plots [34]. As shown in Fig. S15 and Fig. 3g, the band gaps of SiC@UiO-66-NH₂, SiC, SiC/UiO-66-NH₂, and UiO-66-NH₂ are 2.29, 2.32, 2.33, and 2.40 eV, respectively. The slightly narrowed band gap of SiC@UiO-66-NH₂ indicates that the intimate coupling between SiC and UiO-66-NH₂ can modulate the electronic structure and improve the utilization of incident light. This enhanced light harvesting is favorable for generating more photogenerated charge carriers during photocatalysis, thereby providing a prerequisite for efficient TC degradation.

Electrochemical impedance spectroscopy (EIS) was then used to probe the interfacial charge transfer resistance, because a smaller semicircle radius in the Nyquist plot generally corresponds to faster charge transfer across the electrode or catalyst interface [35]. As shown in Fig. 3h, SiC@UiO-66-NH₂ exhibits the smallest semicircle radius among all samples, demonstrating the lowest interfacial charge transfer resistance. This result indicates that the Si–O–Zr linked interface formed by in situ growth can facilitate charge migration between SiC and UiO-66-NH₂ more effectively than the loose contact in the physically mixed SiC/UiO-66-NH₂ sample. Therefore, the chemically bonded heterointerface plays a crucial role in accelerating interfacial electron transfer and reducing kinetic barriers during the photocatalytic process.

Photoluminescence spectroscopy (PL) was further employed to

evaluate the recombination behavior of photogenerated electron–hole pairs, since the PL emission intensity is generally positively correlated with the radiative recombination rate of charge carriers. [36] As shown in Fig. 3i, SiC@UiO-66-NH₂ shows the weakest PL intensity, suggesting that the recombination of photogenerated electrons and holes is significantly suppressed. In contrast, the stronger PL emission of the individual components and the physically mixed sample indicates less efficient charge separation. These PL results are consistent with the photocurrent and EIS analyses, collectively confirming that the in situ formed SiC@UiO-66-NH₂ heterostructure can simultaneously enhance charge generation, promote interfacial charge transfer, and inhibit carrier recombination. Therefore, the superior photocatalytic performance of SiC@UiO-66-NH₂ can be attributed to the synergistic effects of improved light absorption and efficient charge separation enabled by the high-quality chemically bonded heterointerface.

DFT calculations were further performed to elucidate the interfacial charge regulation effect of the SiC@UiO-66-NH₂ heterostructure at the atomic scale. Charge density difference analysis is a powerful method for visualizing electron accumulation and depletion after heterointerface formation [37]. As shown in Fig. 4a and b, obvious electron redistribution occurs at the SiC/UiO-66-NH₂ interface, with electrons mainly transferring from Si atoms in SiC to O atoms in UiO-66-NH₂. To quantitatively evaluate the interfacial charge transfer, Bader charge analysis was further conducted (Fig. S16), revealing that approximately 10.6 electrons are transferred from SiC to UiO-66-NH₂ in the SiC@UiO-66-NH₂ heterostructure. This directional electron redistribution indicates strong electronic coupling between the two components and confirms that the chemically bonded interface can serve as an effective bridge for interfacial charge migration.

The electronic structure of the heterostructure was further investigated by density of states (DOS) analysis, which provides insight into the band structure evolution after interfacial coupling [38]. As shown in Fig. 4c, the theoretical band gap of SiC@UiO-66-NH₂ is smaller than those of individual SiC and UiO-66-NH₂, which is consistent with the experimental band-gap trend obtained from UV–Vis DRS analysis. The narrowed band gap suggests that the formation of the SiC/UiO-66-NH₂ heterostructure can modulate the electronic structure and improve the photoresponse, thereby facilitating the generation of photogenerated charge carriers under light irradiation.

Electrostatic potential calculations were then used to clarify the origin of the built-in electric field at the heterointerface. As shown in Fig. 4d, the vacuum levels on the SiC side and UiO-66-NH₂ side of SiC@UiO-66-NH₂ are 2.23 and 2.15 eV, respectively. The potential difference between the two sides demonstrates the formation of a built-in electric field at the interface, which provides an internal driving force for directional charge migration. Combined with the electrostatic potential distributions of the individual SiC and UiO-66-NH₂ models (Fig. S17), this result further verifies that the interfacial potential gradient originates from the contact between the two components.

Work function and Fermi level analyses were further conducted to understand the driving force for charge redistribution. As shown in Fig. 4e, the different Fermi levels of SiC and UiO-66-NH₂ lead to electron redistribution after contact until interfacial equilibrium is reached. This process induces band bending and establishes a built-in electric field at the chemically bonded interface. Therefore, the Si–O–Zr linked SiC@UiO-66-NH₂ heterostructure can promote directional charge transfer, suppress electron–hole recombination, and provide a theoretical basis for its enhanced photocatalytic performance.

The extended dark-adsorption measurements provide direct experimental evidence for TC uptake (Fig. S10). The equilibrium adsorption capacities of UiO-66-NH₂, SiC, SiC@UiO-66-NH₂, and physically mixed SiC/UiO-66-NH₂ were 4.96, 2.97, 5.45, and 3.92 mg g⁻¹, respectively, with SiC@UiO-66-NH₂ showing the highest capacity. This trend is consistent with its large BET surface area and distinct pore structure, as well as with the DFT adsorption energies of –44.6, –24.3, and –17.4 eV for TC on SiC@UiO-66-NH₂, UiO-66-NH₂, and SiC, respectively (Fig. 4f).

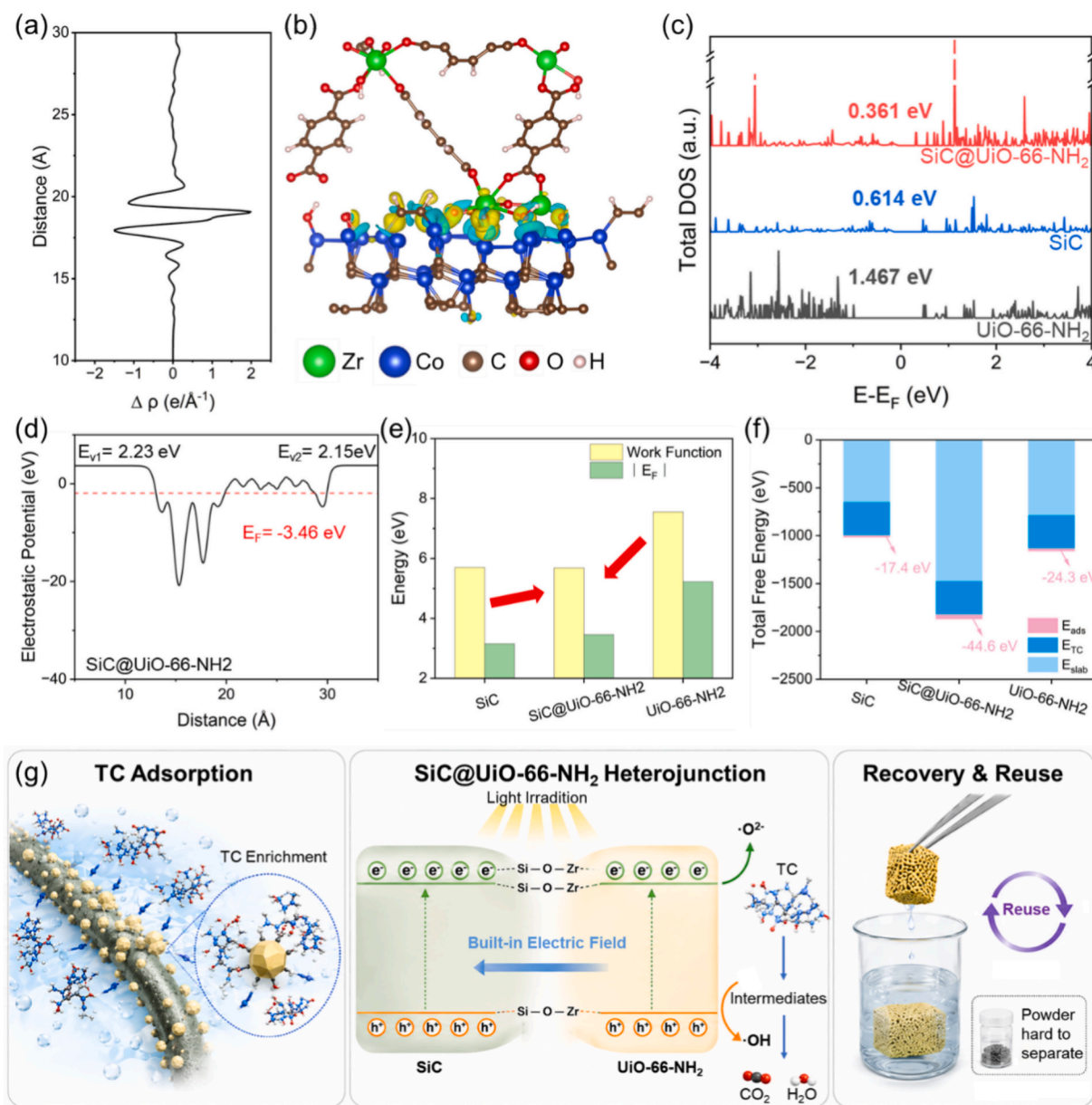


Fig. 4. (a and b) 2-dimensional and 3-dimensional charge density difference plots of SiC@UiO-66-NH₂. (c) Density of states of SiC@UiO-66-NH₂, SiC, and UiO-66-NH₂. (d) Electrostatic potential distribution of SiC@UiO-66-NH₂. (e) Work functions and Fermi levels of SiC@UiO-66-NH₂, SiC, and UiO-66-NH₂. (f) Adsorption energies of TC on SiC@UiO-66-NH₂, SiC, and UiO-66-NH₂. (g) Schematic illustration of the photocatalytic enhancement mechanism over SiC@UiO-66-NH₂.

These mutually consistent experimental and theoretical results support the enhanced TC adsorption/enrichment ability of the in situ constructed composite.

Nevertheless, the modest adsorption-capacity difference between SiC@UiO-66-NH₂ and UiO-66-NH₂ cannot by itself account for the pronounced photocatalytic advantage of the composite. Adsorption is therefore regarded as an enhanced but cooperative factor, whereas the superior activity is mainly attributed to its synergistic coupling with improved interfacial charge separation induced by the chemically bonded Si–O–Zr heterointerface. Based on the above results, the proposed mechanism (Fig. 4g) combines pollutant enrichment with interface-driven charge redistribution, a built-in electric field, accelerated charge migration, and suppressed electron–hole recombination.

The time-dependent evolution of TC and its intermediates was evaluated using a normalized LC-MS peak-area heatmap constructed from samples collected at 0, 20, 40, 60, 90, and 120 min (Fig. 5a). [39] The relative peak area of TC continuously decreased and reached

approximately 0.08 at 120 min. P1 and P2 were mainly formed during the early stage and reached relatively high levels at about 40 min before declining. P3–P6 reached their maxima mainly during 60–90 min, whereas the low-molecular-weight products P7–P10 were generated mainly in the later stage, reached relatively high levels around 90 min, and then decreased to different extents at 120 min. Thus, most intermediates were subsequently degraded rather than being permanently accumulated in solution.

To further elucidate the photocatalytic degradation process of TC over SiC@UiO-66-NH₂, LC-MS was employed to identify the intermediates generated during the reaction, and possible degradation pathways were proposed based on the detected mass-to-charge signals (Fig. 5b). The characteristic peak of the parent TC molecule was observed at $m/z = 445$. As the photocatalytic reaction proceeded, the TC molecule was gradually attacked by reactive species and underwent successive reactions, including hydroxylation, demethylation, deamination, decarboxylation, ring opening, and molecular fragmentation.

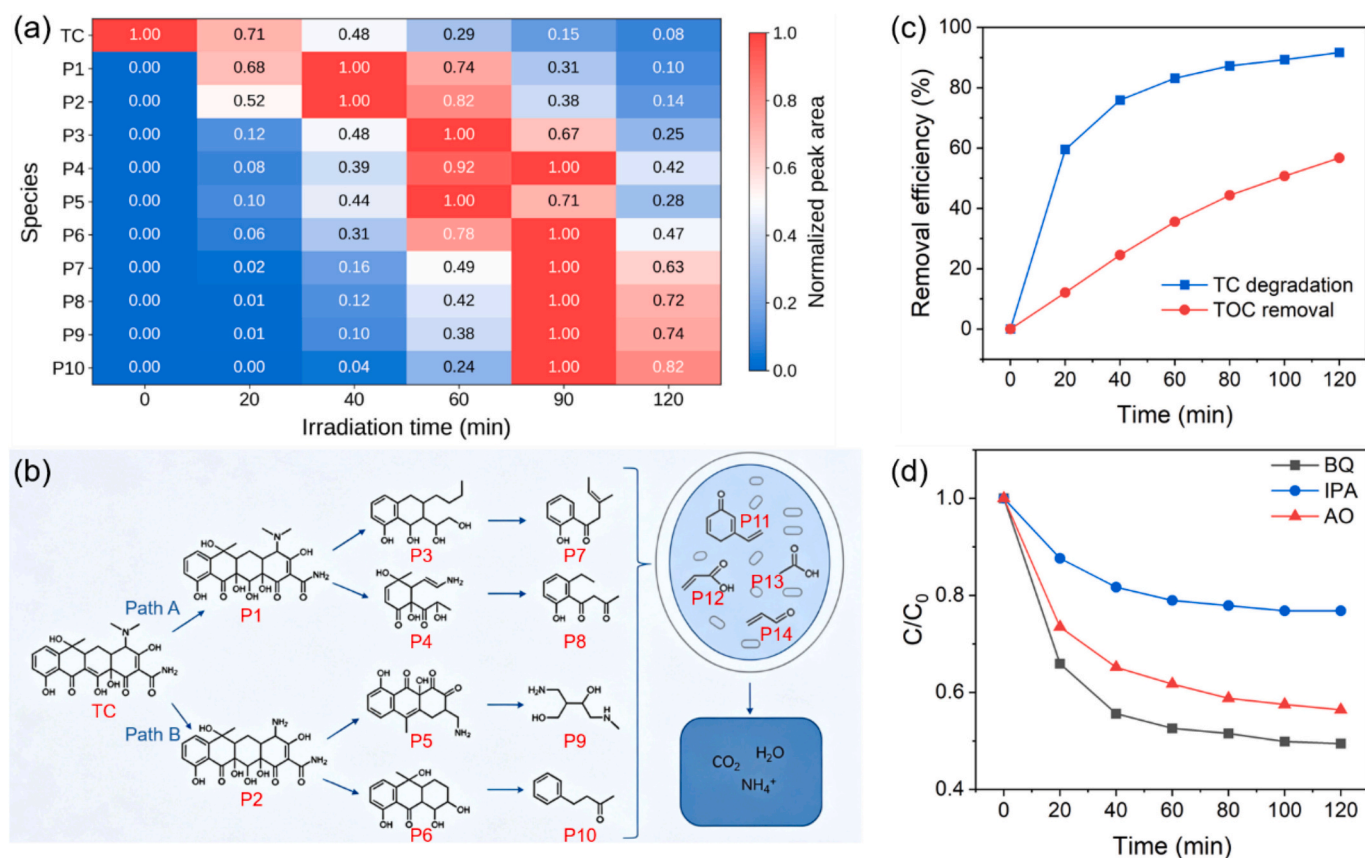


Fig. 5. (a) Normalized LC-MS peak-area heatmap showing the time-dependent evolution of TC and intermediates P1–P10 during photocatalytic degradation, (b) Proposed photocatalytic degradation pathways of TC based on LC-MS analysis, (c) Comparison of TC degradation efficiency and TOC removal efficiency as a function of irradiation time, and (d) Effect of different scavengers on the photocatalytic degradation of TC over SiC@UiO-66-NH₂.

These processes ultimately led to the formation of low-molecular-weight products, which could be further mineralized into inorganic species such as CO₂ and H₂O.

According to the LC-MS results, the degradation of TC in the SiC@UiO-66-NH₂ system mainly follows two possible pathways [40,41]. In pathway I, reactive species first attack the C–C bond in the inner ring of the TC molecule, triggering hydroxylation and generating intermediate P1 ($m/z = 461$) [42]. Subsequently, P1 undergoes deamination, partial hydroxylation, and carbonylation, accompanied by ring cleavage to form P3 ($m/z = 280$) and P4 ($m/z = 256$). Further removal of carbonyl or carboxyl groups, together with additional ring-opening reactions, produces smaller intermediates such as P7 ($m/z = 190$) and P8 ($m/z = 205$). In pathway II, demethylation of the tertiary amine group in TC first occurs, yielding P2 ($m/z = 433$). This intermediate then undergoes deamination, partial hydroxylation, ring opening, and C–C bond cleavage, leading to the formation of P5 ($m/z = 301$) and P6 ($m/z = 279$). After further removal of hydroxyl and carbonyl groups, the molecular skeleton continues to break down, producing low-molecular-weight intermediates such as P9 and P10 ($m/z = 149$). With prolonged irradiation, the intermediates formed through these two pathways can be continuously oxidized by reactive species, including $\cdot\text{O}_2^-$, $\cdot\text{OH}$, and h^+ , and further fragmented into smaller organic species, such as P11–P14 ($m/z = 121, 73, 60, \text{ and } 57$). Eventually, these low-molecular-weight products are further oxidized and mineralized into inorganic products such as CO₂, H₂O, and NH₄⁺. These results indicate that SiC@UiO-66-NH₂ can effectively destroy the parent structure of TC and promote the deep oxidation of its intermediates, providing evidence for the reduction of TC residues and their potential ecological risks [43].

TOC analysis was performed to evaluate mineralization (Fig. 5c). [44] After 20 min, the TC degradation efficiency had reached 59.49%,

whereas TOC removal was only 12.09%, indicating rapid destruction of the parent molecular structure before extensive mineralization. At 40, 60, and 80 min, TC degradation increased to 75.91%, 83.17%, and 87.26%, while TOC removal rose to 24.52%, 35.61%, and 44.35%, respectively. After 120 min, TC degradation reached 91.65% and TOC removal reached 56.78%. Together with the time-resolved LC-MS results, these data confirm a stepwise process in which TC is first transformed into intermediates that are subsequently further oxidized and partially mineralized, although some organic intermediates remain after 120 min.

Scavenger experiments were further used to identify the reactive species involved in TC degradation (Fig. 5d). After 120 min, the C/C_0 values in the presence of BQ, IPA, and AO were 0.4945, 0.7680, and 0.5643, corresponding to TC degradation efficiencies of 50.55%, 23.20%, and 43.57%, respectively, compared with 91.65% without scavengers. The strongest inhibition caused by IPA indicates that $\cdot\text{OH}$ is the predominant reactive species, while the inhibition observed with AO and BQ demonstrates that h^+ and $\cdot\text{O}_2^-$ also make appreciable contributions.

4. Conclusions

In summary, a monolithic SiC@UiO-66-NH₂ composite aerogel with chemically bonded Si–O–Zr heterointerfaces was successfully constructed by the in situ growth of UiO-66-NH₂ on hydroxylated SiC nanofiber aerogels. Under Xe-lamp irradiation, the composite achieved 91.65% TC degradation within 120 min, with a single apparent pseudo-first-order rate constant of 0.01516 min^{-1} obtained from the 20–120 min fitting interval, outperforming pristine SiC, UiO-66-NH₂, and physically mixed SiC/UiO-66-NH₂. Blank and extended dark-adsorption

tests confirmed negligible direct photolysis and an equilibrium TC adsorption capacity of 5.45 mg g^{-1} for SiC@UiO-66-NH₂, demonstrating that adsorption enrichment contributes cooperatively to photocatalysis. The aerogel maintained high activity across broad pH and TC-concentration ranges and retained over 90% degradation efficiency and 98.2% mass retention after 15 cycles. ICP, SEM, and XRD analyses further confirmed strong immobilization of UiO-66-NH₂ without obvious pH-dependent leaching or structural deterioration. Time-resolved LC-MS and TOC analyses showed that intermediates were continuously oxidized and that 56.78% TOC removal was achieved after 120 min, while scavenger experiments identified $\cdot\text{OH}$ as the predominant reactive species, with h^+ and $\cdot\text{O}_2^-$ also contributing. Overall, the Si–O–Zr heterointerface promotes directional charge transfer, establishes a built-in electric field, and suppresses carrier recombination, while the porous composite facilitates pollutant enrichment and monolithic recovery. This work provides a feasible strategy for designing efficient, stable, and recyclable MOF/ceramic aerogel photocatalysts for hazardous antibiotic wastewater remediation.

CRedit authorship contribution statement

Han Zhang: Writing – original draft. **Yu Wang:** Data curation. **Jiayi Zhao:** Data curation. **Danchen Sun:** Data curation. **Lei Zuo:** Data curation. **Encan Kang:** Formal analysis. **Jincan Che:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jphotochem.2026.117635>.

Data availability

Data will be made available on request.

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